

Ali AGHABABAEI-HARANDI

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[Google Scholar](#)

Summary

Final-year Ph.D. candidate at Université Grenoble Alpes, supervised by Professor Massih-Reza Amini. Research focuses on model efficiency and representation learning within machine learning and deep learning, with the goal of designing scalable and practical AI systems.

Education

Université Grenoble Alpes

2023 – Present

Ph.D. in Computer Science

- **Research:** Deep Learning Efficiency, Representation Learning
- Supervised by Professor Massih-Reza Amini, APTIKAL team

Sharif University of Technology

2018 – 2022

M.Sc. in Computer Science

- **Coursework:** Deep Learning, Speech Processing

University of Tehran

2013 – 2018

B.Sc. in Electrical Engineering

- **Coursework:** Linear Algebra, Pattern Recognition, Digital Signal Processing

Research Interests

- **Deep Learning Efficiency:** Tensor Decomposition, Knowledge Distillation, Memory-Efficient Training
- **Representation Learning:** Semi-Supervised Learning, Pseudo-Labeling, Contrastive Learning
- **Generative AI:** Probabilistic Modeling, Image Denoising, Image Super-Resolution

Research Experience

Ph.D. Thesis

2023 – Present

APTIKAL team, Université Grenoble Alpes

- Working on a **memory-efficient** training method that improves greedy supervised learning by preserving information during local training via neural information estimation.
- Working on extending the **Forward-Forward** algorithm to Vision Transformers through early cross-attention between label tokens and feature representations.
- Investigated semi-supervised learning via **pseudo-labeling** through a theoretical analysis of bias stability; proposed an adaptive **Gaussian geometric** loss to shape latent representations, accelerate convergence, and reduce majority-class bias.
- Reduced computational complexity and parameter count of pre-trained models through a **tensor decomposition** approach with a custom **constraint-based** loss function for optimal rank selection while preserving accuracy.

Research Collaborator

2021 – 2022

University of Basel

- Investigated inverse problems (image denoising, limited-view CT, wave scattering) using invertible neural networks.
- Approximated data distributions with normalizing flows to identify out-of-distribution samples, improving model robustness and reliability.

M.Sc. Thesis

2018 – 2021

Electronic Research Institute, Sharif University of Technology

- Proposed a patch-wise feature analysis approach for identifying forgery in video frames, enhancing deepfake detection capabilities.
- Developed a method to improve generalization and robustness of deepfake detection models.

B.Sc. Thesis

2016 – 2017

University of Tehran

- Designed and implemented a no-reference video quality meter to assess blockiness and blurriness distortions.

Publications

JEPAMatch: Geometric Representation Shaping for Semi-Supervised Learning

March 2026

Ali Aghababaei-Harandi, Aude Sportisse, Massih-Reza Amini

ECML-PKDD 2026 — [\[link\]](#)

Unified Framework for Pre-trained Neural Network Compression via Decomposition and Optimized Rank Selection

September 2025

Ali Aghababaei-Harandi, Massih-Reza Amini

ECML-PKDD 2025 — [\[link\]](#)

Deep Variational Inverse Scattering

March 2023

AmirEhsan Khorashadizadeh, Ali Aghababaei-Harandi, Tin Vlašić, Hieu Nguyen, Ivan Dokmanić

EuCAP 2023 — [\[link\]](#)

Conditional Injective Flows for Bayesian Imaging

February 2023

AmirEhsan Khorashadizadeh, Konik Kothari, Leonardo Salsi, Ali Aghababaei-Harandi, Maarten de Hoop, Ivan Dokmanić

IEEE Transactions on Computational Imaging — [\[link\]](#)

Talks & Presentations

Efficient Deep Learning

December 2025

Mathematical Fundamentals of AI Workshop, Sorbonne Center for AI ([SCAI](#)), Paris, France

Presentation on Pre-trained Neural Network Compression

September 2025

European Conference on Machine Learning ([ECML-PKDD](#)), Porto, Portugal

Conférence sur l'Apprentissage Automatique ([CAp](#)), Dijon, France

Skills

Programming: Python, PyTorch, TensorFlow, JAX, Hugging Face, Git

Theoretical: Pattern Recognition, Design and Analysis of Algorithms, Deep Learning